U.S. SEMICONDUCTOR: TRADE & LABOR ANALYSIS

Why These Datasets Were Selected and How They Were Collected | 2025

Motivation for the Analysis

The semiconductor industry is central to global competitiveness. Inflationary pressures, labor costs, tariffs, and international demand are reshaping supply chains and influencing strategic decisions for governments, CEOs, startups, and recruiters. This analysis was designed to connect trade flows, labor trends, and logistics data into one coherent picture — identifying nearshoring opportunities for the U.S. and its partners, quantifying risks from tariff structures and concentrated value chains, and demonstrating the ability to collect, clean, and analyze complex datasets for actionable insights.

Why These Datasets

All datasets selected are official, legal, and publicly available, ensuring credibility and reproducibility. Each source was chosen to answer a specific dimension of the problem.

Dataset	Source	Why It Was Chosen	Analytical Purpose
Trade Flows (Imports/Exports)	UN Comtrade (HS 85423100)	Global standard for trade statistics	Compare U.S. semiconductor imports/exports by partner country
Tariff Structures	WTO Tariff Data, USITC HTS, Section 301 Tariffs	Authoritative tariff schedules	Quantify competitiveness impacts of MFN vs. Section 301 penalties
Labor & Employment	BLS (QCEW, ECI, SAE, OEWS)	Comprehensive U.S. labor statistics	Assess employment costs, wage trends, and talent availability
Logistics & Transportation	Rail Equipment Accident Data (Form 54)	Operational efficiency and risk	Evaluate cost savings and reliability in trade routes
Global Demand & Market Trends	OECD, SEMI.org, SIA	Industry-level demand projections	Forecast growth and identify emerging hubs

Why This Analysis

This analysis bridges trade, labor, and logistics because these three pillars jointly determine competitiveness. Trade flows show where goods move. Tariffs reveal policy barriers and incentives. Labor data highlights costs and talent pipelines. Logistics data uncovers efficiency and risk in supply chains. Demand projections provide forward-looking context. By combining them, the analysis delivers insights that are strategic, actionable, and relevant to executives, recruiters, and policymakers.

How the Data Was Collected

- **Phase 1:** Located and extracted datasets from official portals (UN Comtrade, WTO, USITC, BLS, OECD, SEMI).

- **Phase 2:** Downloaded raw files (CSV, XLS, PDF) and API outputs, ensuring proper documentation of each source.
- **Phase 3:** Cleaned and standardized millions of rows, aligning HS codes, NAICS categories, and time series formats.
- **Validation:** Cross-checked figures across multiple sources (e.g., WTO vs. USITC) to ensure consistency.
- **Documentation:** Each dataset is accompanied by a PDF or README in the repository for full transparency.

Why From These Sources

- **Credibility:** All sources are internationally recognized institutions (UN, WTO, OECD, BLS).
- **Reproducibility:** Publicly available datasets ensure anyone can replicate the analysis.
- **Coverage:** Together, they cover trade, tariffs, labor, logistics, and demand — the full ecosystem.
- **Strategic Value:** These sources provide the kind of data CEOs, recruiters, and policymakers rely on for decision-making.

Project Structure

This project is structured in five phases: data collection, cleaning, EDA, visualization, and reporting. The analytical phases and their interdependencies are summarized below.

Phase	Title	Status	Description
1	Data Collection	Complete	Official datasets sourced from UN, WTO, OECD, BLS
2	Data Cleaning	Complete	Normalization, type conversion, deduplication
3	EDA	In Progress	Statistical exploration, pattern identification
4	Visualization	In Progress	Power BI, Python charts, Excel dashboards
5	Reporting	Upcoming	Strategic findings and recommendations

Document Metadata

- **Data Sources:** UN Comtrade, WTO, USITC, BLS, OECD, SEMI.org, SIA
- **HS Code:** 85423100 — Electronic integrated circuits (processors and controllers)
- **Repository:** [GitHub — Semiconductor Trade & Labor Analysis — USA 2025]